

TRI-CITIES, TN, USA

WMO: 723350

Lat: 36.480N

Lon: 82.399W

Elev: 1497

StdP: 13.92

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13877

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.8	17.4	1.1	6.1	19.4	5.8	7.8	22.2	23.3	42.3	20.4	39.8	4.2	250	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	90.5	71.9	88.4	71.4	86.5	70.9	75.1	85.3	74.1	83.8	73.1	82.3	6.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.2	126.2	78.8	71.2	121.9	77.7	70.3	117.8	76.9	39.5	85.4	38.5	84.1	37.6	82.1	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.4	13.5	5.2	93.4	6.9	3.2	0.2	95.7	-3.9	97.5	-7.8	99.3	-12.8	101.7		
(5)			4.2	77.0	6.7	1.1	-0.7	77.7	-4.6	78.4	-8.4	79.0	-13.3	79.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.7	36.0	40.1	47.5	56.7	64.9	72.3	75.1	74.5	68.7	57.4	46.4	39.6		
	DBStd	15.83	10.75	9.89	9.24	7.93	6.63	4.42	3.73	3.76	5.76	8.00	8.48	9.31		
	HDD50	1476	450	301	158	30	2	0	0	0	0	26	166	343		
	HDD65	4131	899	699	544	263	86	6	0	1	31	256	559	787		
	CDD50	3915	16	22	80	231	463	670	780	760	560	255	57	21		
	CDD65	1095	0	0	1	14	82	226	315	295	141	20	1	0		
	CDH74	9230	0	5	24	222	821	1863	2509	2360	1195	221	10	0		
	CDH80	3048	0	0	1	28	184	621	927	864	385	38	0	0		
(14) Wind	WSAvg	4.2	5.2	5.3	5.8	5.4	4.3	3.6	3.3	2.8	2.8	3.3	3.8	4.3		
(15) Precipitation	PrecAvg	40.6	3.2	3.4	3.9	3.3	3.8	3.7	4.6	3.2	3.0	2.4	2.9	3.5		
	PrecMax	51.8	5.3	6.3	9.2	7.0	8.7	7.4	9.1	7.1	7.1	5.3	6.7	6.8		
	PrecMin	30.7	1.4	0.7	1.3	0.2	1.3	0.7	1.8	0.6	0.5	0.0	0.5	0.2		
	PrecStd	5.2	1.1	1.5	1.8	1.5	1.7	1.6	1.9	1.3	1.7	1.4	1.3	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.0	72.4	77.2	83.4	87.5	91.7	93.4	93.1	90.9	84.0	75.0	69.6		
		MCWB	57.9	59.7	60.4	64.2	68.2	71.3	73.5	73.0	69.0	68.3	61.3	59.5		
	2%	DB	61.8	65.3	72.6	79.7	84.4	88.8	90.1	90.3	87.8	79.6	69.9	62.7		
		MCWB	54.8	54.4	58.1	62.1	67.4	70.8	73.0	72.3	69.4	64.6	57.9	55.2		
	5%	DB	57.1	60.3	68.3	76.4	81.9	86.3	87.9	87.8	84.7	75.8	66.0	58.3		
		MCWB	51.3	51.3	55.5	60.6	66.5	70.4	72.5	71.8	68.7	62.3	56.1	52.8		
	10%	DB	52.8	55.7	63.8	72.7	79.2	83.8	85.5	85.1	81.4	72.2	61.9	54.7		
		MCWB	48.4	48.7	53.1	59.1	65.5	69.6	72.0	71.1	67.4	61.3	53.7	50.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.5	62.1	62.4	67.8	72.3	75.3	77.2	76.1	74.0	71.6	64.7	61.1		
		MCDB	63.8	69.5	73.4	78.9	81.8	85.1	86.9	87.0	84.0	79.8	70.7	66.7		
	2%	WB	56.0	57.2	60.0	64.9	70.3	73.8	75.4	74.8	72.2	68.1	61.2	57.1		
		MCDB	60.7	63.1	69.2	75.6	80.1	84.0	85.4	85.3	81.5	74.8	67.2	60.1		
	5%	WB	52.8	53.4	57.8	62.8	68.7	72.5	74.3	73.7	70.7	65.6	58.3	54.3		
		MCDB	56.2	58.4	65.5	72.3	78.3	82.1	83.8	83.6	79.1	72.5	63.8	57.5		
	10%	WB	48.9	50.1	55.1	60.7	67.1	71.2	73.3	72.7	69.3	63.1	55.0	50.8		
		MCDB	52.2	54.4	61.8	69.1	76.0	80.0	82.2	81.6	77.2	69.8	60.7	54.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.5	19.8	22.4	24.1	22.9	21.3	19.8	21.0	22.5	23.8	22.7	18.8		
		MCDBR	24.3	27.4	29.3	29.8	26.9	25.2	23.5	24.5	27.2	28.6	28.2	24.0		
		MCWBR	17.1	17.7	16.3	14.7	11.6	9.6	8.5	8.8	10.5	14.2	17.0	16.7		
	5% WB	MCDBR	20.4	21.9	23.7	23.8	22.6	20.9	19.9	20.5	20.8	21.6	21.7	19.7		
		MCWBR	15.8	16.4	14.8	12.9	10.6	8.9	8.2	8.1	8.9	12.0	15.2	15.3		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.309	0.334	0.375	0.418	0.446	0.481	0.483	0.382	0.352	0.314	0.294		
	taud		2.517	2.486	2.452	2.349	2.239	2.193	2.128	2.099	2.368	2.447	2.520	2.557		
	Ebn at Noon		286	296	296	288	275	266	256	252	276	276	278	281		
	Edh at Noon		25	29	33	39	44	46	49	49	36	30	25	23		
(44) All-Sky Solar Radiation	RadAvg		705	888	1273	1626	1800	1938	1802	1711	1464	1138	838	618		
	RadStd		54	113	104	123	132	119	117	76	119	133	88	59		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.88	N/A	N/A	N/A	N/A	N/A	N/A	-216	+270	+151
(47) Regional (37 neighbors)		+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+257	+158

Nomenclature: See separate page